

Hanoi University of Science and Technology

Faculty of Mathematics and Informatics

CALCULUS I EXERCISE

COURSE ID: MI 1016

1.1-1.3. Functions. Essential functions

Exercise 1. Determine the domain of the following functions.

a) $y = \frac{x}{\sqrt{4x^2 - 1}}$. ✓✓

c) $y = \ln \frac{1-x}{1+x}$. ✓✓

b) $y = \arcsin \frac{x}{x+2}$. ✓✓

d) $y = \sqrt{\arctan x}$. ✓✓

Exercise 2. Find the range of the following functions.

a) $y = \ln(1 - 2\sin x)$. ✓✓

c) $y = \sqrt{\arccos x}$. ✓

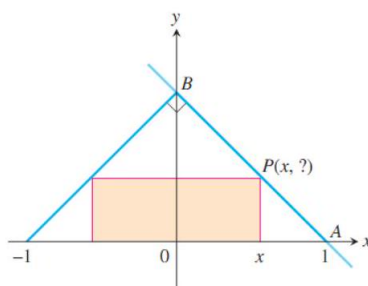
b) $y = \arctan(2e^x)$. ✓✓

d) $y = \frac{x^2 - 1}{x^2 + 1}$. ✓✓

✓ **Exercise 3.** As dry air moves upward, it expands and cools. The ground temperature is 30°C and the temperature at a height of 1 km is 20°C .a) Express the temperature T (in $^\circ\text{C}$) as a function of the height h (in kilometers), assuming that a linear model is appropriate.

b) Draw the graph of the function.

c) What is the temperature at a height of 4 km?

✓✓ **Exercise 4.** The figure shown here shows a rectangle inscribed in an isosceles right triangle whose hypotenuse is 2 units long.

- a) Express the y -coordinate of P in terms of x .
b) Express the area of the rectangle in terms of x .

Exercise 5. Determine whether f is even, odd, or neither.

✓a) $f(x) = \frac{e^x - e^{-x}}{2} =: \sinh x$.

d) $f(x) = \ln \frac{1-x}{1+x}$.

b) $f(x) = \frac{2^x - x^2}{2^x + x^2}$.

e) $f(x) = \sin x + \cos 2x$.

c) $f(x) = \ln(x + \sqrt{x^2 + 1})$.

✓f) $f(x) = \sin x + \sin 2x$.

Exercise 6. Find the functions $f \circ g, g \circ f, f \circ f$, and $g \circ g$ and their domains.

a) $f(x) = \frac{1}{x+1}, g(x) = x-1$.

c) $f(x) = \sin x, g(x) = \sqrt{x+1}$.

b) $f(x) = \sqrt{2x+3}, g(x) = x^2+1$.

d) $f(x) = 1-x^2, g(x) = \frac{1-x}{1+x}$.

Exercise 7. Find the inverse functions of the following functions.

a) $y = \arcsin 2x$.

✓c) $y = \frac{1-3x}{1+3x}$.

b) $y = \frac{e^{2x} - e^{-2x}}{2}$.

d) $y = \ln \frac{e^x - 1}{e^x + 1}$.

2.1 - 2.3. Limits

Exercise 8. Find the limit of the following sequences (if it exists).

a) $u_n = \sqrt[3]{n^3 + 2n^2} - n.$

d) $u_n = \frac{n}{n^2 + 1} + \frac{n}{n^2 + 2} + \cdots + \frac{n}{n^2 + n}.$

b) $u_n = \tan\left(\frac{2n\pi}{1 + 8n}\right).$

e) $u_n = \left(1 - \frac{1}{2n}\right)^n.$

c) $u_n = \cos\left(\frac{n\pi}{2}\right).$

f) $u_n = \frac{n \cos(n^2 + 1)}{n^2 + 2}.$

Exercise 9. Find the limit of the sequence $\{\sqrt{2}, \sqrt{2 + \sqrt{2}}, \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \dots\}.$

Exercise 10. Determine the function α which is infinite as $x \rightarrow \infty$ and the integer n (if there exists) such that α and x^n are of the same order.

a) $\alpha(x) = x^3 + 2x^2 + x^5.$

c) $\alpha(x) = \sin(x^2).$

b) $\alpha(x) = 3x^4 + \sin x.$

d) $\alpha(x) = e^x + x^2.$

Exercise 11. Determine the function α which is infinitesimal as $x \rightarrow 0$ and the integer n (if there exists) such that α and x^n are of the same order.

a) $\alpha(x) = x^3 + 2x^2 + x^5.$

c) $\alpha(x) = \sin(x^2).$

b) $\alpha(x) = 3x^4 + \sin x.$

d) $\alpha(x) = e^x + x^2.$

Exercise 12. Compare the order of the following infinitesimals as $x \rightarrow 0$.

a) $\alpha(x) = \sin(x^2 + x), \beta(x) = 1 - \cos x.$

c) $\alpha(x) = e^{\sin x} - 1, \beta(x) = \arcsin(\tan x).$

b) $\alpha(x) = \sqrt{x^3 + 2x^4}, \beta(x) = \ln(1 + 2x),$
 $x \rightarrow 0^+.$

d) $\alpha(x) = \tan(-x^2 + 3x^3), \beta(x) = \sinh(3x^2).$

Exercise 13. Evaluate the following limits.

a) $\lim_{x \rightarrow 1} \frac{\sqrt{x^2 + 8} - 3}{x - 1}.$

d) $\lim_{x \rightarrow 1} \frac{x^6 - 1}{x^{10} - 1}.$

b) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}.$

c) $\lim_{x \rightarrow 0} \frac{9^x - 5^x}{4^x - 3^x}.$

e) $\lim_{x \rightarrow 0} \frac{\ln(1 + 2x^2)}{1 - \cos x}.$

$$\text{f) } \lim_{x \rightarrow \infty} \left(\frac{x+1}{x+3} \right)^{2x}.$$

$$\text{i) } \lim_{x \rightarrow 0^+} \frac{\sqrt{x^5}}{\sqrt{\sin x} \ln(1-3x^2)}.$$

$$\text{g) } \lim_{x \rightarrow \infty} \left(\sin \frac{1}{x} + \cos \frac{1}{x} \right)^x.$$

$$\text{j) } \lim_{x \rightarrow 0} \frac{\cos(\sin x) - 1}{\sin(\cos x - 1)}.$$

$$\text{h) } \lim_{x \rightarrow 0} \frac{\ln(1+3\tan x)}{e^x - \cos x}.$$

Exercise 14. If $\lim_{x \rightarrow 1} \frac{f(x) - 8}{x - 1} = 9$, find $\lim_{x \rightarrow 1} f(x)$.

3.1 - 3.6. Continuity of functions

Exercise 15. For what value of a is $f(x) = \begin{cases} x^2 + 2, & \text{if } x < 1 \\ 2ax^3 + 1, & \text{if } x \geq 1 \end{cases}$ continuous at every x ?

Exercise 16. Show that f is continuous on $(-\infty, \infty)$.

$$\text{a) } f(x) = \begin{cases} \sin x & \text{if } x < \frac{\pi}{4} \\ \cos x & \text{if } x \geq \frac{\pi}{4} \end{cases}.$$

$$\text{b) } f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ \sqrt{x} & \text{if } x \geq 1 \end{cases}.$$

Exercise 17. Locate the discontinuity of the function and illustrate by graphing

$$\text{a) } y = \frac{1}{1 + e^{1/x}}.$$

$$\text{b) } y = \ln(\tan^2 x).$$

$$\text{c) } y = \frac{\sin x}{2^x - 1}.$$

Exercise 18. Find the numbers at which f is discontinuous. At which of these numbers is f continuous from the right, from the left, or neither?

$$\text{a) } f(x) = \begin{cases} \frac{2^x - 1}{x} & \text{if } x < 0 \\ 2x + c & \text{if } x \geq 0 \end{cases}.$$

$$\text{b) } f(x) = \begin{cases} \frac{\sin^2(\pi x)}{\ln(1+2x^2)} & \text{if } x < 1 \\ \sqrt{x} & \text{if } x \geq 1 \end{cases}.$$

Exercise 19. Prove that there is a root of the given equation in the specified interval.

$$\text{a) } x^6 - 3x + 1 = 0, \quad (0, 1).$$

$$\text{b) } x^3 = \sqrt{3x+1}, \quad (1, 2).$$

Exercise 20. A train starts at 8AM from Hanoi to Haiphong, arriving at 11AM. The next day it starts at 8AM from Haiphong to Hanoi, arriving at 11AM. Is there a point on the route the train will cross at exactly the same time of day on both days?

Exercise 21. Let f be a continuous function on a close interval $[0, 1]$ and $f(0) = 1, f(1) = 0$. Prove that there is a number $c \in (0, 1)$ at which $f(c) = c$.

4.1 - 4.6. Derivatives

Exercise 22. Find the derivative of the following functions:

a) $y = (x^2 + 1) \sqrt[3]{x^2 + 2}$.

d) $y = \ln(x + \sqrt{x^2 + 5})$.

b) $y = \sin(\tan x)$.

e) $y = \sin^n x \cos nx$.

c) $y = \sqrt{x + \sqrt{x}}$.

f) $y = \left(1 + \frac{1}{x}\right)^x$.

Exercise 23. For what values of a and b will

$$f(x) = \begin{cases} ax & \text{if } x < 2 \\ ax^2 + bx + 3 & \text{if } x \geq 2 \end{cases}$$

be differentiable for all values of x ? Discuss the geometry of the resulting graph of f .

Exercise 24. Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ mx + b & \text{if } x > 2 \end{cases}$. Find the values of m and b that make f differentiable everywhere.

Exercise 25. Let $r(x) = f(g(h(x)))$, where $h(1) = 2$, $g(2) = 3$, $h'(1) = 4$, $g'(2) = 5$, and $f'(3) = 6$. Find $r'(1)$.

Exercise 26. If $F(x) = f(3f(4f(x)))$, where $f(0) = 0$ and $f'(0) = 2$, find $F'(0)$.

Exercise 27. Is the derivative of

$$h(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

continuous at $x = 0$? How about the derivative of $k(x) = xh(x)$? Give reasons for your answer.

Exercise 28. Find the n - th derivatives of the following functions:

a) $y = \frac{1}{x^2 + x}$.

d) $y = \ln(2x^2 + x)$.

b) $y = \frac{x}{x^2 - 4}$.

e) $y = (2x + 1) \cos 3x$.

c) $y = (x^2 + 1)e^{2x}$.

f) $y = \cos(2x) \sin x$.

Exercise 29. If f and g are differentiable functions with $f(0) = g(0) = 0$ and $g'(0) \neq 0$, show that

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{f'(0)}{g'(0)}.$$

Exercise 30. Prove each of the following.

- a) The derivative of an even function is an odd function.
- b) The derivative of an odd function is an even function.

Exercise 31. Find the derivative of the function $f(x) = \begin{cases} x \arctan \frac{1}{x^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$.

Exercise 32. Suppose that the functions f and g are defined throughout an open interval containing the points x_0 , that f is differentiable at x_0 , that $f(x_0) = 0$, and that g is continuous at x_0 . Show that the product fg is differentiable at x_0 .

5.1 - 5.5. Applications of Derivatives and Differentials

Exercise 33. Use a linear approximation (or differentials) to estimate the given number.

- a) $\sqrt{1.002}$.
- c) $\sin(0.01)$.
- b) $\sqrt[3]{8.001}$.
- d) $(1.999)^4$.

Exercise 34. Find equations of the tangent line and the normal line to the curve

- a) $y = \ln(x + \sqrt{x^2 + 3})$ at $x = 1$.
- b) $y = x + \tanh(2x)$ at $x = 0$.

Exercise 35. Find a cubic function $y = ax^3 + bx^2 + cx + d$ whose graph has horizontal tangents at the points $(-2, 6)$ and $(2, 0)$.

Exercise 36. Find all the point on the curve $y = 2x^3 - 3x^2 - 12x + 20$ where the tangent of the curve at such a point is

- a) perpendicular to the line $y = 1 - \frac{x}{24}$,
- b) parallel to the line $y = \sqrt{2} - 12x$.

Exercise 37. Show that the tangents to the curve $y = \frac{\pi \sin x}{x}$ at $x = \pi$ and $x = -\pi$ intersect at right angles.

Exercise 38. Given $f(x) = \ln \frac{x+1}{x+2}$, find $df(x)$, $d^{10}f(x)$.

Exercise 39. Given $f(x) = (x + 2) \ln x$, find $d^2 f(1)$, $d^{20} f(1)$.

Exercise 40. Find the n th-degree Taylor polynomials centered at $x = 0$ of $f(x)$. Determine the remainder.

a) $f(x) = x \cos x, n = 5$.

c) $f(x) = \sqrt{2 + 2x}, n = 3$.

b) $f(x) = \frac{x}{\sqrt{1 + x^2}}, n = 5$.

d) $f(x) = e^{2x} + 1, n = 4$.

5.6 - 5.9. Applications of Derivatives and Differentials

Exercise 41. Evaluate the following limits

a) $\lim_{x \rightarrow 0} \frac{\arcsin x - x}{x^3}$.

g) $\lim_{x \rightarrow -\infty} (x^2 + 2^x) \frac{1}{x}$.

b) $\lim_{x \rightarrow 0} \frac{x^5 - \ln(1 + x^5)}{\sin^{10} x}$.

h) $\lim_{x \rightarrow 0} [\ln(e + 2x)] \frac{1}{\sin x}$.

c) $\lim_{x \rightarrow 0} x \ln |x|$.

i) $\lim_{x \rightarrow 0} [2x + e^{3x}] \frac{1}{\sin x}$.

d) $\lim_{x \rightarrow +\infty} x[\pi - 2 \arctan(3x)]$.

j) $\lim_{x \rightarrow 0^+} [\arcsin 2x]^{\tan x}$.

e) $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$.

k) $\lim_{x \rightarrow 0} \frac{\sin x \ln(x+1) - x^2}{x^3}$.

f) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{e^{2x} - 1} \right)$.

l) $\lim_{x \rightarrow 0^+} x^{\sin x}$.

Exercise 42. Show that

- a) $\sin(\arccos x) = \cos(\arcsin x) = \sqrt{1 - x^2}$ for all $x \in [-1, 1]$.
- b) $\frac{1}{2} \arctan \frac{2x}{1 - x^2} = \arctan x$ for all $x \in (-1, 1)$.
- c) $\arcsin(\tanh x) = \arctan(\sinh x)$.

Exercise 43. Prove that

- a) $|\arcsin x - \arcsin y| \geq |x - y|$ for all $x, y \in [-1, 1]$.
- b) $\frac{y - x}{1 + y^2} < \arctan y - \arctan x < \frac{y - x}{1 + x^2}$ for all $0 < x < y$.
- c) $\frac{1}{2} - \frac{x}{8} < \frac{1}{x} - \frac{1}{e^x - 1} < \frac{1}{2}$ for all $x > 0$.
- d) $\frac{x}{x + 1} \leq \ln(x + 1) \leq x$ for all $x > -1$.

Exercise 44. Show that the equation $3x + 2 \cos x + 5 = 0$ has exactly one real root.

Exercise 45. Show that the equation $a \cos x + b \cos 2x + c \cos 3x = 0$ has at least one root on $(0, \pi)$.

Exercise 46. Suppose that $f(x)$ is a continuous function on a close interval $[a, b]$ and differentiable on (a, b) , and $f(a) = f(b) = 0$. Show that there exists a number $c \in (a, b)$ such that $f'(c) = 2021f(c)$.

Exercise 47. For what values of a and b is the following equation true?

$$\lim_{x \rightarrow 0} \left(\frac{\sin 2x}{x^3} + a + \frac{b}{x^2} \right) = 0.$$

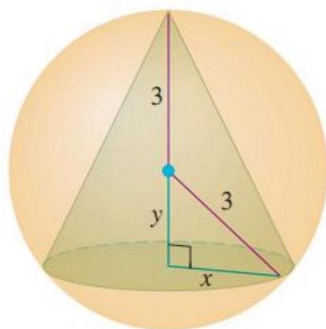
Exercise 48. Determine the local extreme values

- a) $y = x^{2/3}(x + 2)$.
- b) $y = x^{2/3}(x^2 - 4)$.
- c) $y = x\sqrt{4 - x^2}$.
- d) $y = x^2 \ln x$.
- e) $y = (x^2 - 3)e^x$.
- f) $y = 3 \arctan x - \ln(x^2 + 1)$.
- g) $y = \ln(x + 3) + \operatorname{arccot} x$.

Exercise 49. Find the absolute maximum and minimum values of $f(x) = x^2 + \frac{250}{x}$ over $[1, 10]$.

Exercise 50. (The best fencing plan) A rectangular plot of farmland will be bounded on one side by a river and on the other three sides by a single-strand electric fence. With 800 m of wire at your disposal, what is the largest area you can enclose, and what are its dimensions?

Exercise 51. Find the volume of the largest right circular cone that can be inscribed in a sphere of radius 3.



Exercise 52. (Designing a can) What are the dimensions of the lightest open-top right circular cylindrical can that will hold a volume of 1000 cm^3 .

Exercise 53. A rectangle is to be inscribed under the arch of the curve $y = 4 \cos(x/2)$ from $x = -\pi$ to $x = \pi$. What are the dimensions of the rectangle with largest area, and what is the largest area?

Exercise 54. Find the dimensions of a right circular cylinder of maximum volume that can be inscribed in a sphere of radius 10 cm. What is the maximum volume?

Exercise 55. Determine the asymptotes of the graph of $y = f(x)$.

a) $y = \frac{e^x}{x+1}$.

c) $y = (x+2)e^{1/x}$.

d) $y = \sqrt[3]{x^3 + x}$.

b) $y = x \operatorname{arccot} \frac{2}{x}$.

e) $y = e^x \ln x$.

Exercise 56. Determine the asymptotes of the curves.

a) $x = t^3 - 3\pi, y = t^3 - 6 \arctan t$.

b) $x = \frac{t^2}{t-1}, y = \frac{t}{t^2-1}$.

Exercise 57. If f' is continuous, $f(2) = 0$, and $f'(2) = 7$, evaluate

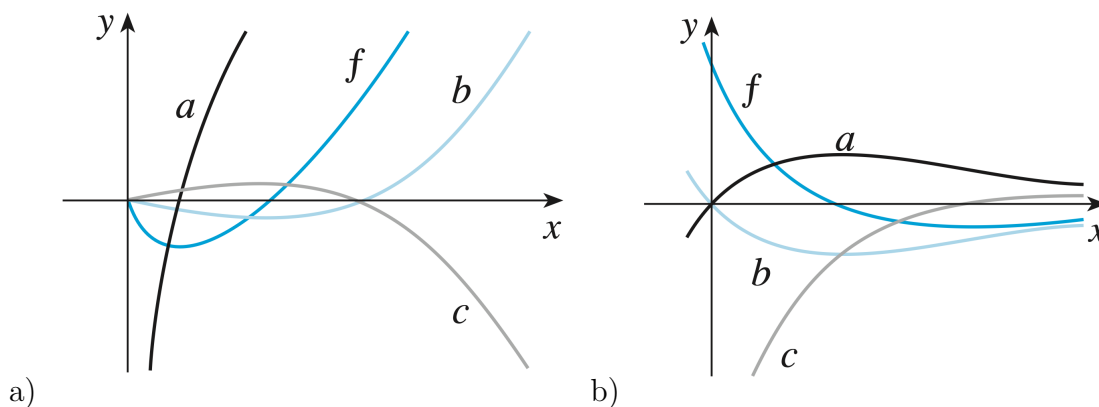
$$\lim_{x \rightarrow 0} \frac{f(2+3x) - f(2+5x)}{x}.$$

6.1 - 6.5. Indefinite Integrals 1

Exercise 58. Find the function f .

- a $f'(x) = 1 + x, f(0) = 1$.
- b $f'(x) = 5x^4 - 3x^2 + 4, f(-1) = 2$.
- c $f''(x) = -2 - 12x^2, f(0) = 4, f'(0) = 12$.
- d $f''(x) = 20x^3 + 12x^2, f(0) = 0, f'(0) = 1$.

Exercise 59. The graph of a function f is shown. Which one (a, b or c) is the anti-derivative of f ? Give your explanation.



Exercise 60. What constant acceleration is required to increase the speed of a car from 30 km/h to 50 km/h in 5s?

Exercise 61. Find a function $f(x)$ such that $f'(x) = x^3$ and the line $x + y = 0$ is tangent to the graph of $f(x)$.

Exercise 62. Evaluate the following integrals

a) $\int x \sin(x^2) dx.$

e) $\int x \sin x dx.$

b) $\int \frac{x+1}{x^2+2x+2} dx.$

f) $\int x^2 e^x dx.$

c) $\int \frac{1}{x \ln^2 x} dx.$

g) $\int \tan 2x dx.$

d) $\int \frac{x}{\sqrt{x+1}} dx.$

h) $\int e^x \sin x dx.$

6.6 - 6.8. Indefinite Integrals 2

Exercise 63. Evaluate the following integrals

a) $\int \frac{x^3+1}{x^2+4} dx.$

e) $\int \frac{\sin 2x}{\sqrt{\sin^4 x + 1}} dx.$

b) $\int \tan^4 x dx.$

f) $\int \frac{dx}{3 \sin x - 4 \cos x}.$

c) $\int \frac{1-2x}{\sqrt{2+x^2}} dx.$

g) $\int \frac{dx}{1 + \sqrt{x^2 + 4x + 5}}.$

d) $\int \frac{x}{(x^2+1)(x+2)} dx.$

h) $\int \frac{x+1}{\sqrt{x^2-2x-1}} dx.$

Exercise 64. Evaluate the following integrals

a) $\int (x+1) \arctan x dx.$

g) $\int \sqrt{\frac{x}{x-1}} dx.$

b) $\int (x+2) \ln x dx.$

h) $\int \frac{x^2+2}{x^3-1} dx.$

c) $\int \arcsin^2 x dx.$

i) $\int \frac{x^2+1}{x^4+1} dx.$

d) $\int \frac{\arctan x}{x^2} dx.$

j) $\int \frac{\sin^2 x}{\cos^3 x} dx.$

e) $\int \frac{x}{(x^2+2x+2)^2} dx.$

k) $\int \frac{1}{x^2 \sqrt{x^2+1}} dx.$

f) $\int \frac{e^{2x}}{1+e^x} dx.$

7.1 - 7.5. Definite Integrals 1

Exercise 65. Show that

a) $\frac{13}{42} < \int_0^1 \sin(x^2) dx < \frac{1}{3}.$

b) $\int_0^{\pi/6} \cos(x^2) dx > \frac{1}{2}.$

Exercise 66. Find the derivative of the following functions.

a) $f(x) = \int_0^x \sqrt{1+t^4} dt.$

c) $h(x) = \int_{x^3}^0 \sin^3 t dt.$

b) $g(x) = \int_0^{x^2} \sin(t^2) dt.$

d) $k(x) = \int_{x^2}^{x^3} \arcsin(t) dt.$

Exercise 67. Evaluate the limit by first recognizing the sum as a Riemann sum for a function defined on $[0, 1]$.

a) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^4}{n^5}.$

b) $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{1 + \frac{1}{n}} + \sqrt{1 + \frac{2}{n}} + \cdots + \sqrt{1 + \frac{n}{n}} \right).$

Exercise 68. Evaluate the following integrals.

a) $\int_1^e (x \ln x)^2 dx.$

c) $\int_1^2 \frac{\sqrt{x^2 - 1}}{x^2} dx.$

b) $\int_0^{\pi/4} \frac{\sin^2 x \cos x}{(1 + \tan^2 x)^2} dx.$

d) $\int_0^1 \frac{\ln(x^2 + 1)}{(x + 1)^2} dx.$

Exercise 69. If f is continuous on $[0, 1]$, show that

a) $\int_0^{\pi/2} f(\sin x) dx = \int_0^{\pi/2} f(\cos x) dx.$

b) $\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$

Exercise 70. Evaluate

a) $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx.$

b) $\int_0^{\pi/2} \frac{\sqrt[3]{\sin x}}{\sqrt[3]{\sin x} + \sqrt[3]{\cos x}} dx.$

7.6 - 7.8. Definite Integrals 2

Exercise 71. Find the area of the region enclosed by the parabolas $x = 2y - y^2$, $x = y^2 - 4y$.

Exercise 72. Find the area of the region enclosed by the curve $y^2 = x^2 - x^4$.

Exercise 73. Find the area of the region enclosed by $y = \frac{1}{x}$, $y = x$ and $y = \frac{1}{4}x$, $x > 0$.

Exercise 74. Find the number b such that the line $y = b$ divides the region bounded by the curves $y = x^2$ and $y = 4$ into two regions with equal area.

Exercise 75. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

- a) $y = 2x - x^2$, $y = 0$; about the x -axis. c) $x = y^2$, $x = 1$; about $x = 1$.
b) $y = \ln x$, $y = 1$, $y = 2$, $x = 0$; about
the y -axis. d) $y = x^2$, $x = y^2$; about $y = -1$.

Exercise 76. Find the volume of the solid generated by revolving the region bounded on the left by the parabola $x = y^2 + 1$ and on the right by the line $x = 5$ about

- a) the x -axis.
b) the y -axis.
c) the line $x = 5$.

Exercise 77. Find the length of the curves

- a) $y = \frac{x^2}{8} - \ln x$, $4 \leq x \leq 8$.
b) $x = y^{2/3}$, $1 \leq y \leq 8$.
c) $x = 5 \cos t - \cos 5t$, $y = 5 \sin t - \sin 5t$, $0 \leq t \leq \pi/2$.

Exercise 78. Find the area of the surface generated by revolving the curve

- a) $y = \sqrt{x^2 + 2}$, $0 \leq x \leq \sqrt{2}$, about the x -axis.
b) $y = \frac{1}{4}x^2 - \frac{1}{2} \ln x$, $1 \leq x \leq 2$, about the y -axis.

7.9 - 7.10. Definite Integrals 3

Exercise 79. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_0^\infty \frac{x}{(x^2 + 2)^2} dx.$

f) $\int_0^\infty \frac{e^x}{e^{2x} + 3} dx.$

b) $\int_1^\infty \frac{x + 2}{x^2 + 3x} dx.$

g) $\int_0^\infty \frac{x \arctan x}{(1 + x^2)^2} dx.$

c) $\int_{-\infty}^\infty \frac{x}{x^2 + 1} dx.$

h) $\int_1^\infty \frac{x + 1}{\sqrt{x^4 - x}} dx.$

d) $\int_{-\infty}^0 x e^{-x} dx.$

i) $\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx.$

e) $\int_0^1 \frac{\ln x}{\sqrt{x}} dx.$

Exercise 80. Determine whether the improper integral is convergent or divergent.

a) $\int_e^\infty \frac{1}{x(\ln x)^p} dx.$

e) $\int_0^1 \frac{dx}{x - \sin x}.$

b) $\int_1^\infty \frac{dx}{\sqrt{x + x^3}}.$

f) $\int_0^\infty (\sqrt[3]{x^3 + 1} - x) dx.$

c) $\int_1^\infty \frac{\sin x}{x^2 + x + 1} dx.$

g) $\int_0^\infty \frac{\sin x}{x} dx.$

d) $\int_0^1 \frac{\sqrt{x} dx}{\sqrt{1 - x^4}}.$

h) $\int_0^\infty \frac{\cos x - \cos 3x}{x^2 \ln(1 + \sqrt{x})} dx.$

Exercise 81. Find the value of the constant C for which the integral $\int_0^\infty \left(\frac{x}{x^2 + 1} - \frac{C}{3x + 1} \right) dx$ converges. Evaluate the integral for this value of C .

Exercise 82. Suppose f is continuous on $[0, \infty)$ and $\lim_{x \rightarrow \infty} f(x) = 1$. Is it possible that $\int_0^\infty f(x) dx$ is convergent?

8.1 - 8.3. Functions of Several Variables 1

Exercise 83. Find and sketch the domain of the function.

a) $f(x, y) = \sqrt{1 - x^2} - \sqrt{4 - y^2}.$

c) $f(x, y) = \frac{\sqrt{y - x^2}}{x^2 - 1}.$

b) $f(x, y) = \arcsin(x^2 + y^2 - 2).$

d) $f(x, y) = \sqrt{x - y} \ln(x + y).$

Exercise 84. Find the domain and range of the function $f(x, y) = \sqrt{4 - x^2 - y^2}.$

Exercise 85. Find the limit, if exists, or show that the limit does not exist.

a) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{1 + x^2 + y^2} - 1}.$

d) $\lim_{(x,y) \rightarrow (\infty, \infty)} \frac{y^2}{x^2 + 3xy}.$

b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{2x^2 + y^4}.$

e) $\lim_{(x,y) \rightarrow (0,0)} \frac{x(e^{2y} - 1) - 2y(e^x - 1)}{x^2 + y^2}.$

c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin y}{3x^2 + y^2}.$

f) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 \cos y + y^3 \cos x}{x^2 + y^2}.$

Exercise 86. For what value of a is $f(x, y) = \begin{cases} x \arctan \frac{1}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ a, & \text{if } (x, y) = (0, 0) \end{cases}$ continuous at every (x, y) ?

Exercise 87. Find the first partial derivatives of the function.

a) $z = \sin \left(\frac{x}{1 + xy} \right).$

d) $z = x^2 \sin \frac{x}{y}.$

b) $z = (x^2 + 1)^y.$

e) $z = \arctan \frac{x}{\sqrt{x^2 + y^2}}.$

c) $z = \int_y^{x^2} \sin(t^2) dt.$

f) $u = x^2 y \arcsin(y + z).$

Exercise 88. Find $\partial z / \partial x$ and $\partial z / \partial y$.

a) $z = e^u \sin(uv)$, where $u = xy^2, v = x^2 y.$

b) $z = \arcsin(u - v)$, where $u = x^2 + y^2, v = 1 - 2xy.$

8.4 - 8.6. Functions of Several Variables 2

Exercise 89. Find the second partial derivatives of the function.

a) $f(x, y) = \arctan \frac{y}{x}.$

b) $f(x, y) = \frac{xy}{x - y}.$

c) $f(x, y) = x \ln(x^2 + y^2).$

Exercise 90. Verify that the function $u(x, t) = \sin(x + at)$ satisfies the wave equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}.$$

Exercise 91. Show that the function $u = \sin x \cosh y + \cos x \sinh y$ is a solution of Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Exercise 92. Let $f(x, y) = \begin{cases} \frac{x^3 y - xy^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$

a) Find $f_x(x, y)$ and $f_y(x, y)$.

b) Show that $f_{xy}(0, 0) = -1$ and $f_{yx}(0, 0) = 1$.

Exercise 93. Find the linear approximation of the function $f(x, y) = \sqrt{20 - x^2 - 7y^3}$ at $(2, 1)$ and use it to approximate $f(1.98, 1.05)$.

Exercise 94. Find the differential of the function.

a) $z = x^2 \ln(x + y^2)$.

c) $z = xy e^{xy}$.

b) $z = \arctan \frac{y}{x}$.

d) $z = xy + \sinh(xy)$.

8.7 - 8.9. Functions of Several Variables 3

Exercise 95. Find the $d^2 f(x, y)$.

a) $f(x, y) = x^2 y + y^2 x, \quad (x, y) = (1, 1)$.

b) $f(x, y) = \sin(xy) e^x, \quad (x, y) = (0, 1)$.

c) $f(x, y, z) = x^2 + y^3 + z^4, \quad (x, y) = (-1, 0, 1)$.

d) $f(x, y, z) = \ln(1 + xyz), \quad (x, y, z) = (0, 0, 0)$.

Exercise 96. Use the Chain Rule to find dz/dt .

a) $z = \sqrt{1 + x^4 + y^2}, x = \ln t, y = \sin t$.

b) $z = \cos(x + 2y), x = 3t^2, y = 1/t$.

c) $z = xy + yw + xw, x = \sin t, y = \cos t, w = \tan t$.

d) $z = \frac{x + y}{y - w}, x = t^2, y = t^3, w = t^4$.

Exercise 97. Use the Chain Rule to find $\frac{\partial z}{\partial r}, \frac{\partial z}{\partial s}$.

a) $z = \cos(1 + xy), \quad x = r^2, y = s^2.$

b) $z = x^2 y^3, \quad x = r \cos s, y = r \sin s.$

c) $z = e^x \sin y, \quad x = rs, y = r + s.$

d) $z = \tan\left(\frac{x}{y}\right), \quad x = r^2 + s^2, y = 2rs.$

Exercise 98. Use the Chain Rule to show that if $z = f(x, y)$ and $x = r \cos \theta, y = r \sin \theta$ then

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} + \frac{1}{r} \frac{\partial z}{\partial r}.$$

Exercise 99. Find the gradient of f .

a) $f(x, y) = e^x \sin y.$

c) $f(x, y) = y^2 e^{xy}.$

b) $f(x, y) = \arctan(xy).$

d) $f(x, y, z) = xe^y + ye^z + ze^x.$

Exercise 100. Find the directional derivative of f in the given direction $\mathbf{v}$.

a) $f(x, y) = e^y \cos x, \mathbf{v} = (1, 1).$

c) $f(x, y) = \sin(x^2 + y^2), \mathbf{v} = (0, 2).$

b) $f(x, y) = \frac{x^2}{x^2 + y^2}, \mathbf{v} = (-1, 1).$

d) $f(x, y) = \sqrt{x^2 + y^2}, \mathbf{v} = (1, -1).$

Exercise 101. Use implicit differentiation to find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

a) $xy = \ln(y + z^2).$

d) $x^3 + y^2 + z^3 + 6xyz = 1.$

b) $x - z = \arctan(yz).$

e) $2x^2y + 4y^2 + x^2z + z^3 = 3.$

c) $\sin(xyz) = x + 2y + 3z^3.$

f) $xyz = \arcsin(x + y + z).$

Exercise 102. Find the equation of the tangent line of the curve $2x^3 + 4y^2 = 6$ at the point $(1, 1)$.

Exercise 103. Find $x'(y)$ if x is defined implicitly as a function of y by the equation $y^2 + 2x^3y + x = 0$. Apply the previous computation to find the equation of the tangent line of the curve $y^2 + 2x^3y + x = 0$ at the point $(1, -1)$.

8.10 - 8.11. Functions of Several Variables 4

Exercise 104. Find the local maximum and minimum values and saddle point(s) of the function.

a) $z = xy^3 - 8x + 12y^2$.

e) $z = e^{2x}(4x^2 - 2xy + y^2)$.

b) $z = x^4 + y^4 - 4xy + 2$.

f) $z = x^4 + y^4 - x^2 - y^2 + 2xy$.

c) $z = 2x^3 + xy^2 + 5x^2 + y^2$.

g) $z = x^2 + 4y^2 - 4xy + 2$.

d) $z = e^y(y^2 - x^2)$.

h) $z = x^2ye^{-x^2-y^2}$.

Exercise 105. Find the maximum and minimum values of the following functions:

a) $z = x^2 + y^2 + xy - 7x - 8y$ in the triangle bounded by the lines $x = 0$, $y = 0$, and $x + y = 6$.

b) $z = 4x^2 - 9y^2$ in the domain bounded by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.